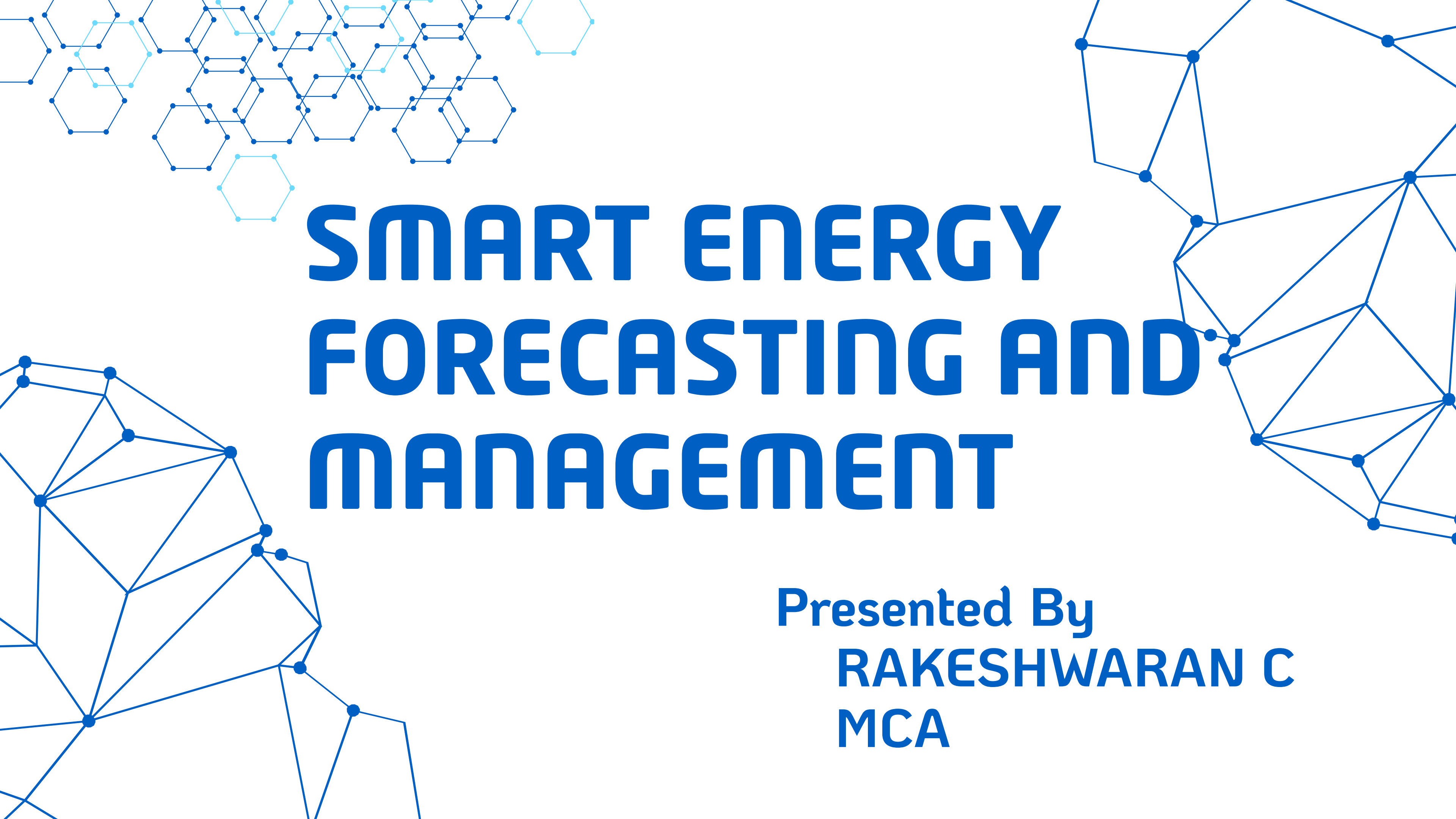
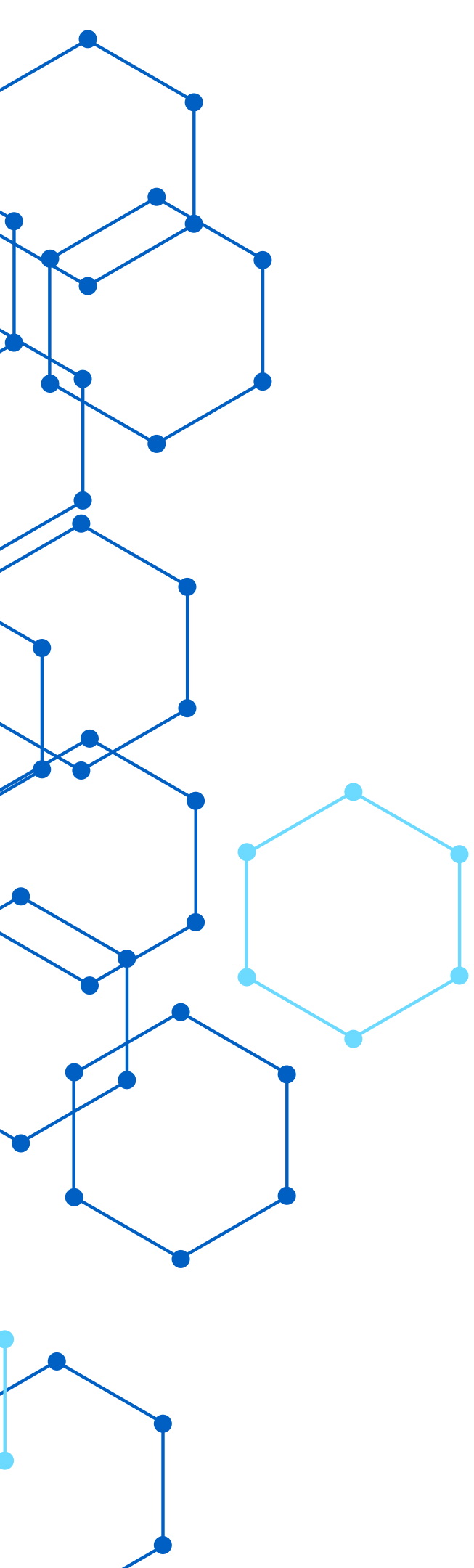


The background features decorative geometric patterns in the corners. The top-left corner has a cluster of interconnected hexagons in blue and light blue. The top-right and bottom-left corners feature complex, interconnected line networks forming irregular polygons. The bottom-right corner has a similar line network pattern.

SMART ENERGY FORECASTING AND MANAGEMENT

**Presented By
RAKESHWARAN C
MCA**



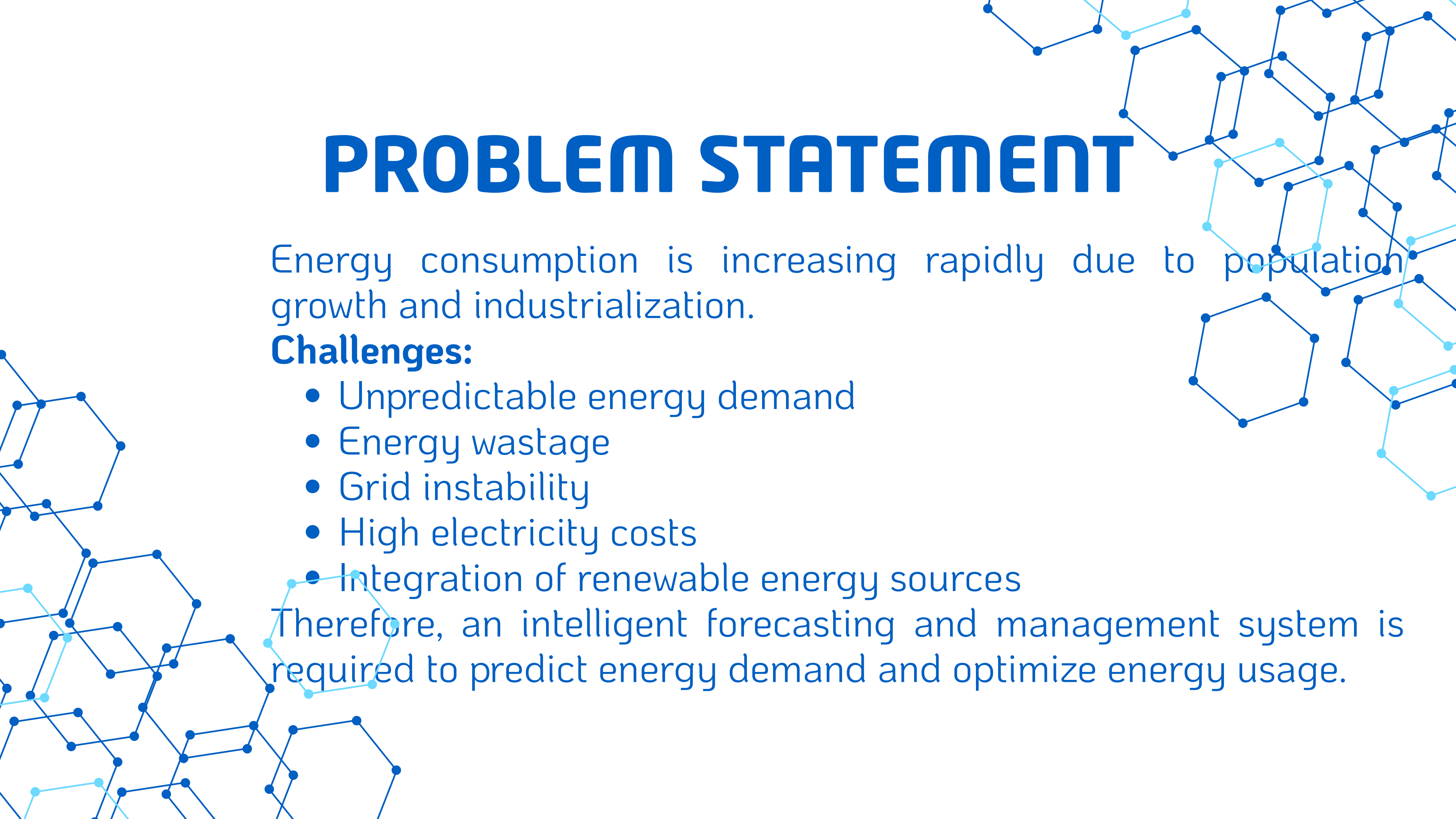
INTRODUCTION

Smart Energy Forecasting and Management is an advanced system that predicts future energy demand and optimizes energy consumption.

It uses:

- Artificial Intelligence (AI)
- Machine Learning (ML)
- Internet of Things (IoT)
- Smart Grid Technologies

These technologies help improve energy efficiency, reduce wastage, and support sustainable development.



PROBLEM STATEMENT

Energy consumption is increasing rapidly due to population growth and industrialization.

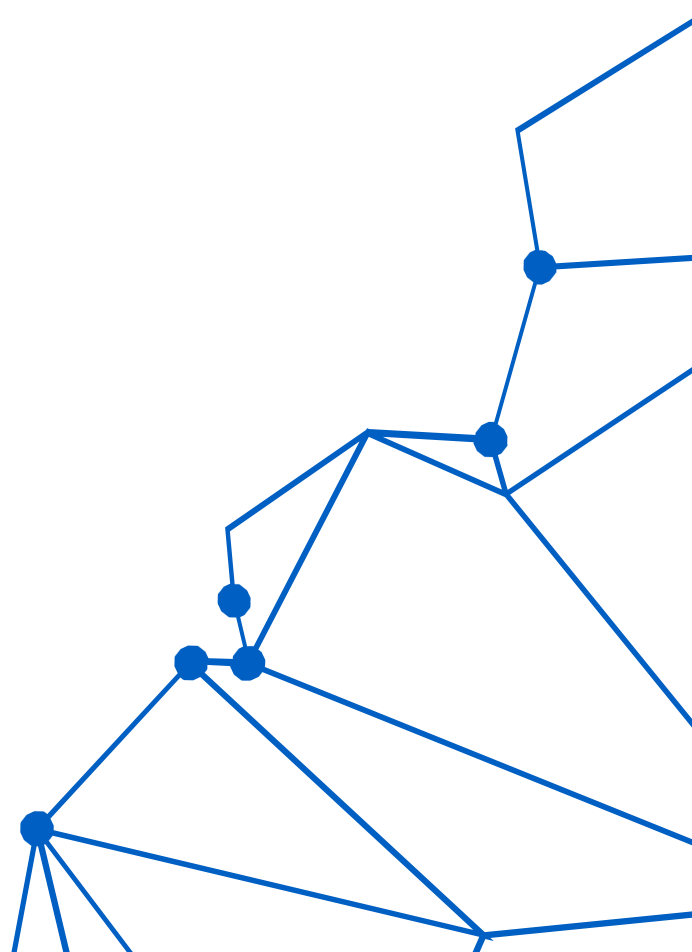
Challenges:

- Unpredictable energy demand
- Energy wastage
- Grid instability
- High electricity costs
- Integration of renewable energy sources

Therefore, an intelligent forecasting and management system is required to predict energy demand and optimize energy usage.



OBJECTIVES

- Predict future energy consumption accurately
 - Analyze energy usage patterns
 - Optimize energy distribution
 - Reduce electricity wastage
 - Support renewable energy integration
 - Improve grid reliability and efficiency
 - Provide real-time monitoring through dashboards
- 

DATASET

Source:

Smart Meter Data / Energy Consumption Dataset

Input Features:

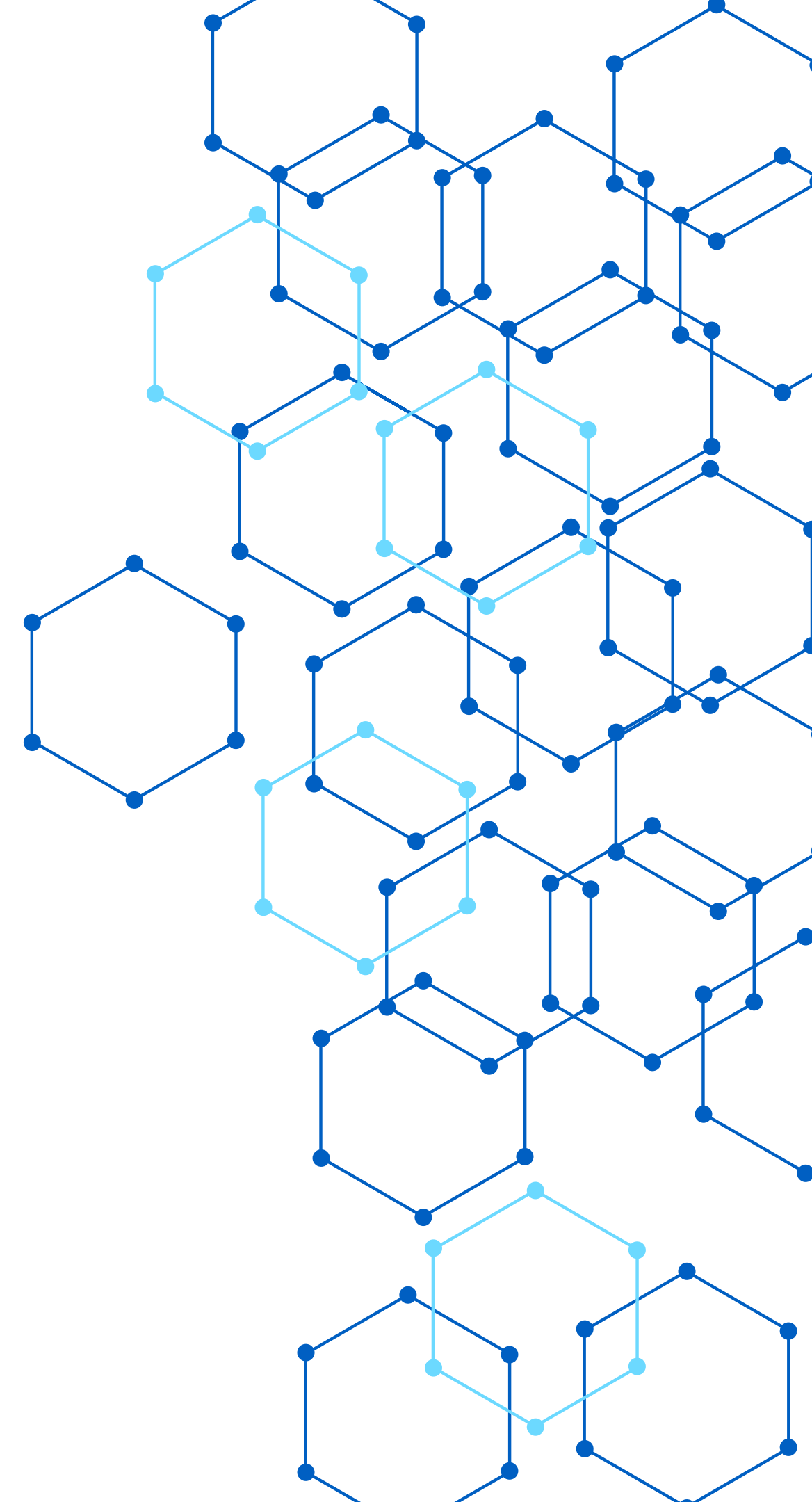
- Temperature
- Humidity

Time and Date

- Energy Consumption History
- Weather Conditions
- Renewable Energy Generation

Target Variable:

- Future Energy Demand
- Energy Consumption Forecast



TECHNOLOGY STACK

Programming Language

- Python

Libraries

- Pandas
- NumPy
- Matplotlib
- Scikit-Learn

Development Environment

- Jupyter Notebook

Visualization

- Power BI Dashboard
- Streamlit Dashboard

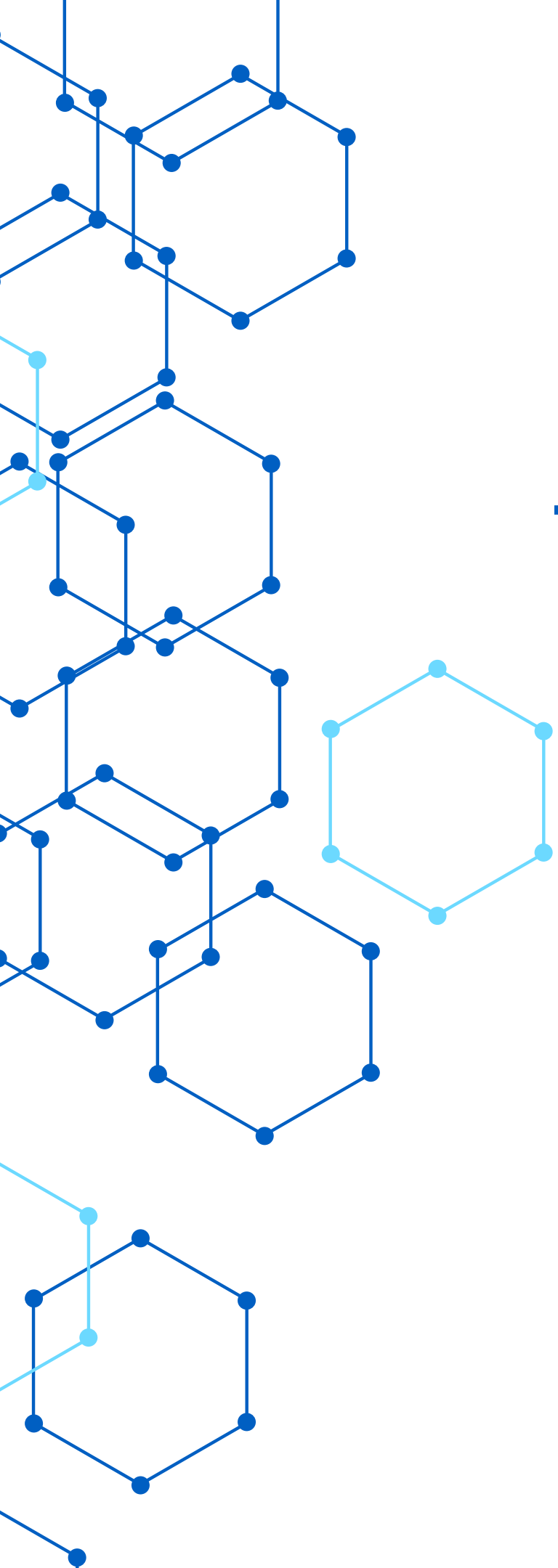


DATA PROCESSING

The collected energy data is cleaned and prepared before training.

Steps Performed:

- Handling Missing Values
- Removing Duplicate Records
- Data Normalization
- Feature Engineering
- Train-Test Data Splitting

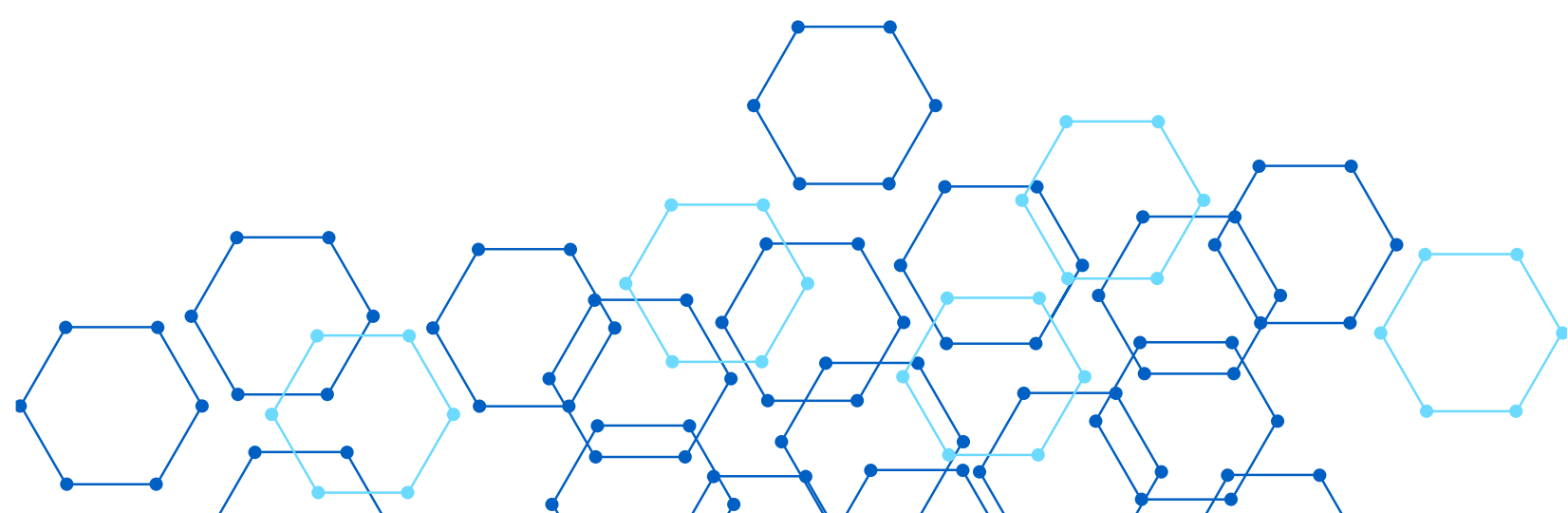


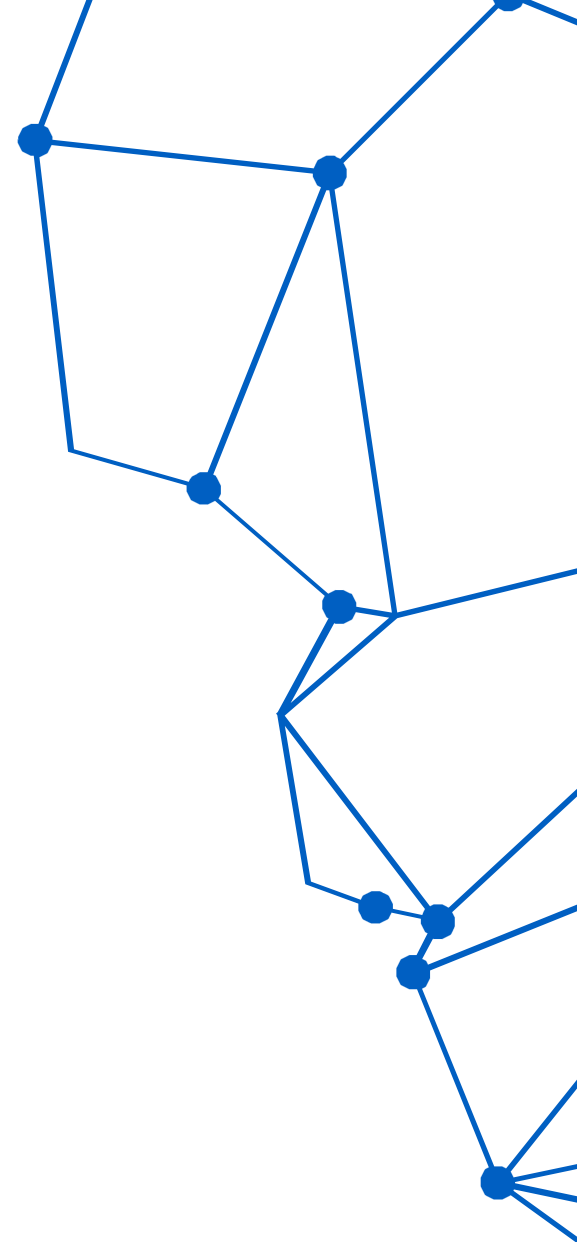
MODEL TRAINING

Training Process

- Dataset divided into:
 - 80% Training Data
 - 20% Testing Data
- Model learns patterns from historical energy consumption data.
- Forecasts future energy demand and consumption trends.

METHODOLOGY

- 
- Data Collection
 - Data Preprocessing
 - Feature Selection
 - Energy Demand Analysis
 - Machine Learning Model Training
 - Energy Forecasting
 - Performance Evaluation
 - Dashboard Visualization
- 



SYSTEM ARCHITECTURE

Smart Meters & IoT Sensors



Data Collection



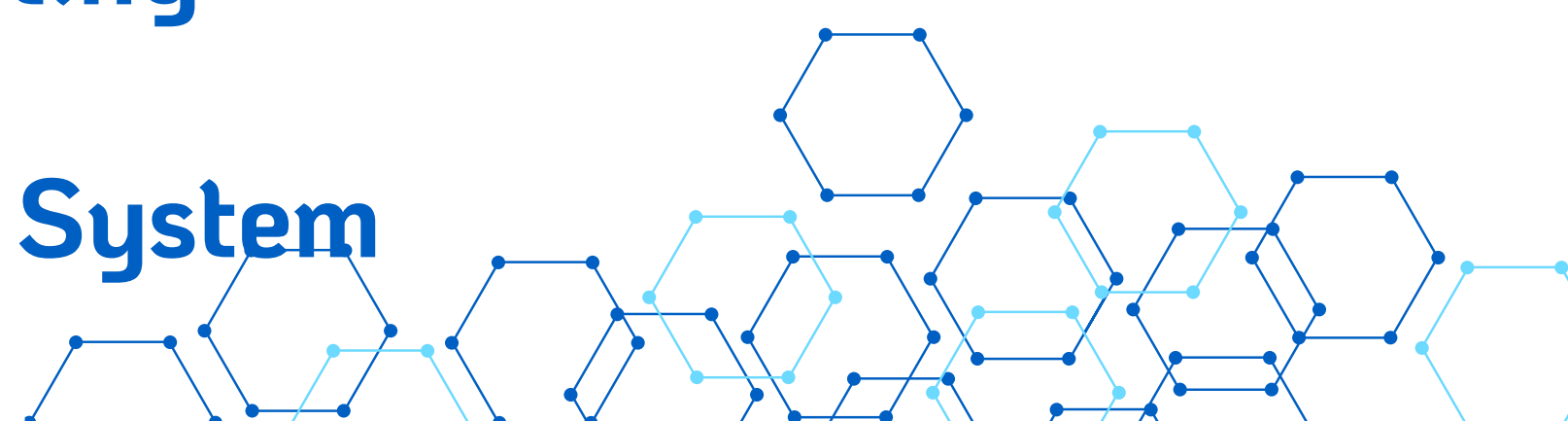
Data Preprocessing



Machine Learning Model



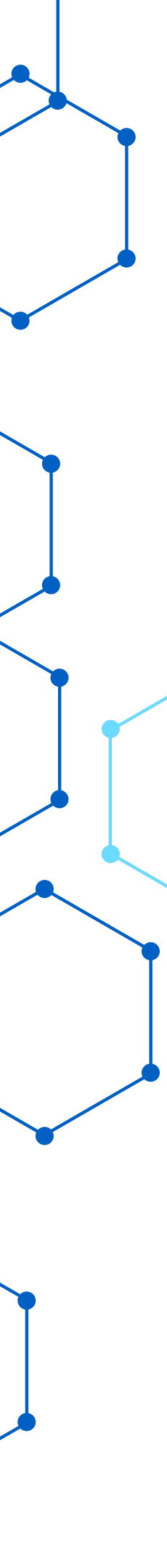
Energy Demand Forecasting



Smart Energy Management System

OUTPUT DASHBOARD





CONCLUSION

The Smart Energy Forecasting and Management System successfully predicts future energy demand using machine learning techniques. The system improves energy efficiency, reduces wastage, supports renewable energy integration, and enhances overall grid performance. It plays a vital role in developing sustainable and intelligent energy management solutions.

The background features a complex network of blue lines and hexagons. On the left, a series of interconnected lines form a web-like structure with several prominent nodes. The top and bottom edges are decorated with a pattern of hexagons, some of which are light blue and others dark blue, creating a textured border. The central area is dominated by the large 'THANK YOU' text.

THANK YOU